SATHYABAMA
INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

TITLE PAGE

- Team Name: The logic lords
- Open Innovation - WattWise
- PS Category- Software
- Team ID-



IDEA TITLE

Problem Statement:

People waste electricity through everyday habits, such as leaving lights, fans, and other appliances running unnecessarily. Most people don't know how much these habits cost them.

Proposed Solution:

WattWise is a website that helps users identify everyday habits that waste electricity and understand how much energy and money they could save by changing them.

Key Features:

- Users record their electricity-saving habits.
- A calculator estimates energy and bill savings.
- WattWise Mystery allows users to identify electricity-wasting habits.
- Power Pact: Lets friends work towards a common goal.
- Energy Impact Garden grows as users complete challenges.

Problem Addressed: This website encourages people to reduce electricity waste and develop better daily habits

Innovation and Uniqueness: Combines electricity cost estimates, interactive challenges, teamwork and a virtual garden in one platform.

TECHNICAL APPROACH

Technologies used:

Frontend: HTML and CSS

Backend: Python with flask

Storage: simple local storage or python data structures

Implementation Process:

- Users record their daily electricity-saving habits
- Python calculates estimated energy and bill savings.
- Points and leaderboards encourage friendly competition.
- Virtual gardens grow as users complete challenges.
- A bill simulator estimates possible future electricity costs.
- **Workflow:**

User input → Python processing → Savings calculation → Points & Results

FEASIBILITY AND VIABILITY

Feasibility of the idea

- Wattwise can be built as a simple website.
- It can be developed using free tools at low cost.
- Users can record daily electricity saving habits.
- Electricity savings can be estimated using appliance power and usage time.

Strategies for overcoming challenges

- Send daily reminders to users.
- Give points, badges , and rewards.
- Keep the website simple and easy to use.
- Use appliance details to estimate savings.

Viability of the idea

- Useful for homes ,schools ,colleges, and offices.
- Makes electricity saving fun through competitions
- Encourages people to follow better electricity saving habits.
- More features can be added in the future.
- Helps people understand how small habits can save electricity.

BUSINESS CANVAS

1. **Mission and Vision**
   - Mission: Encourage people to save electricity through simple daily habits and fun challenges.
   - Vision: Make energy conservation a daily habit through teamwork and technology.

2. **Market and Competitive Analysis**
   - Target Market: Students, colleges, hostels, and households.
   - Market Opportunity: Growing awareness of electricity costs and sustainability.

3. **Business Perspective Analysis**
   - Business Model: Freemium platform.
   - Value Proposition: Make saving electricity fun, simple and measurable.
   - Growth Strategy: Partner with colleges, schools and residential communities.

4. **Financial Planning and Requirements**
   - Requirements: App development, hosting, design and promotion.
   - Revenue: Premium subscriptions, institutional plans and sponsorships.
   - Financial Plan: Start with a low-cost prototype and expand based on user feedback.

IMPACT AND BENEFITS

IMPACT ON USERS:

- Helps students and hostel residents understand their daily electricity usage.
- Encourages people to save electricity through friendly competitions.
- Helps users to see how small daily actions can make difference.

BENEFITS OF WATTWISE:

- **Economic:** It helps to reduce electricity bill by avoiding unnecessary power usage.
- **Environmental:** It saves energy and reduces unnecessary power consumption.
- **Social:** Bring students and households together to achieve electricity-saving goals.
- **Long-term benefit:** Encourages people to build better electricity saving habit.

RESEARCH AND REFERENCES

Research Work:

- Identified common daily habits that lead to electricity wastage.
- Explored simple ways to estimate energy waste and potential cost savings.
- Studied how challenges and teamwork can motivate people to save electricity.

References:

- Mission LiFE – Energy-Saving Actions
  https://missionlife-moefcc.nic.in/List-of-Action.php
- Bureau of Energy Efficiency (BEE), India
  https://beeindia.gov.in/
- Australian Government – Reduce Energy Bills
  https://www.energy.gov.au/households/household-guides/reduce-energy-bills